

Problem Statement Title: Civic Issue Reporting and Resolution System

Local governments often face challenges in promptly identifying, prioritizing, and resolving everyday civic issues like potholes, malfunctioning streetlights, or overflowing trash bins. While citizens may encounter these issues daily, a lack of effective reporting and tracking mechanisms limits municipal responsiveness. A streamlined, mobile-first solution can bridge this gap by empowering community members to submit real-world reports that municipalities can systematically address.

Detailed Description

The system revolves around an easy-to-use mobile interface that allows users to submit reports in real-time. Each report can contain a photo, automatic location tagging, and a short text or voice explanation, providing sufficient context. These submissions populate a centralized dashboard featuring a live, interactive map of the city's reported issues. The system highlights priority areas based on volume of submissions, urgency inferred from user inputs, or other configurable criteria.

On the administrative side, staff access a powerful dashboard where they can view, filter, and categorize incoming reports. Automated routing directs each report to the relevant department such as sanitation or public works based on the issue type and location. System architecture accommodates spikes in reporting, ensuring quick image uploads, responsive performance across devices, and near real-time updates on both mobile and desktop clients.

Expected Solution

The final deliverable should include a mobile platform that supports cross-device functionality and seamless user experience. Citizens must be able to capture issues effortlessly, track the progress of their reports, and receive notifications through each stage — confirmation, acknowledgment, and resolution.

On the back end, a web-based administrative portal should enable municipal staff to filter issues by category, location, or priority, assign tasks, update statuses, and communicate progress. The platform should integrate an automated routing engine that leverages report metadata to correctly allocate tasks to departments.

A scalable, resilient backend must manage high volumes of multimedia content, support concurrent users, and provide APIs for future integrations or extensions. Lastly, the solution should deliver analytics and reporting features that offer insights into reporting trends, departmental response times, and overall system effectiveness — ultimately driving better civic engagement and government accountability.